

APP41339: NetBITS: Future Net-Zero Business-driven Equitable Intelligent Transportation Systems

Application overview

This is what your application will look like.

NetBITS: Future Net-Zero Business-driven Equitable Intelligent Transportation Systems

1. Details

Summary

NetBITS is a groundbreaking model that harnesses power of Digital Twin (DT), Artificial Intelligence (AI), Vehicular Edge Computing (VEC) and Blockchain to revolutionize data exchange in net zero Vehicular Ad hoc Networks (VANETs) and autonomous vehicles within Equitable Intelligent Transportation Systems (ITS). NetBITS innovative approach addresses mobility, equity and security barriers in net zero ITS by taking the data exchange to unprecedented level in which vehicles in ITS can exchange data at different secure mobility levels, thanks to innovative DT and edge-driven solutions. Importantly, NetBITS is the first to facilitate business opportunities models based on DT, AI and Blockchain and VEC, where vehicles and roadside units (RSU) can act

as service providers. In addition, it is worth noting that we embed sustainability and equity as both ethical imperatives and economic drivers, recognizing their significant market potential.

Building on the expertise of the academic and industrial partners the uniqueness of NetBITS will lie in its four-key components:

1. Innovative DT-driven support

NetBITS will develop DT technology to create multi-scale digital mapping of physical objects, and its deployment in VEC can significantly contribute to solving the mobility related problems caused by to the dynamics of ITS due to enormous mobility characteristics of vehicles that change network's topology rapidly and cause intermittent connectivity between a source and its destination.

2. AI-Powered Business Model

The project will enable the ITS to facilitate new business models, in which vehicles and RSU can be service providers. The NetBITS project will build a novel Service Level Agreement framework for net zero mobility, in which data exchange can be facilitated and different parameters are used to meet Quality of Service (QoS) demands. The framework will deploy AI to derive a node hiring policy that can adapt to changes in the system to guarantee the QoS for clients. The business model integrates the incentive and trust mechanisms, allowing vehicles to utilize their data-sharing capabilities to generate revenue. This will pave the way for a wider range of business models and opportunities in net zero mobility-enabled ITS.

3. A novel security solution

NetBITS will fill in the gap caused by a lack of secure adaptive VEC-integrated distributed data exchange scheme and a novel resource management framework that copes with the time-varying network conditions and diverse user requirements in the VEC network. We will develop a native security and privacy for resource-constrained environments, employing blockchain and assuring data privacy in the presence of malicious actors. In conclusion, our expertise in

blockchain technology and cybersecurity in general aligns seamlessly with the proposed project objectives.

4. Visibility and Implementation

In collaboration with our industrial partners CGI, AWTG LTD, and DARE National Hub we will demonstrate our innovative platform's effectiveness through a real-world wireless charging feasibility study, promoting both demonstrate the effectiveness of our solutions. Thus, this will fundamentally revolutionize the telecommunication and computing industries without adding significant investment cost in infrastructure and subsequent efficiency and equitable access. Wireless charging will be considered as a service to operational and maintenance cost, which could potentially offer a disruptive technology in ITS and the wider smart cities domain.

Start date

June 2026

Duration

36 months

2. Core team

Name	Role	Organisation
Ahmed Al-Dubai	Project lead	Edinburgh Napier University
Achille Fonzone	Project co-lead (UK)	Edinburgh Napier University
Amir Hussain	Project co-lead (UK)	Edinburgh Napier University
Baraq Ghaleb	Project co-lead (UK)	Edinburgh Napier University
Berk Canberk	Project co-lead (UK)	Edinburgh Napier

		University
Emma Hart	Project co-lead (UK)	Edinburgh Napier University
Tughrul Arslan	Project co-lead (UK)	University of Edinburgh
Lina Mohjazi	Project co-lead (UK)	University of Glasgow
Mahmoud Wagih	Project co-lead (UK)	University of Glasgow
Muhammad Imran	Project co-lead (UK)	University of Glasgow
Olaoluwa Popoola	Project co-lead (UK)	University of Glasgow

3. Thematic area alignment

Application question

Select the EPSRC thematic area your application most closely aligns to. This should be the theme where the majority of the novelty in the research lies.

In the text box, copy the letter corresponding to your selected theme:

- A. artificial intelligence (AI)
- B. digital security and resilience
- C. energy and decarbonisation
- D. engineering
- E. healthcare technologies
- F. information and communication technologies (ICT)

G. manufacturing

H. mathematical sciences

I. physical sciences

J. quantum technologies

Additional guidance:

This is for administrative purposes to help the initial application processing. We will check your choice and make a final decision on which theme will lead the peer review of your application.

We will not publish closing dates for individual rounds, instead we will batch applications internally. You should apply when your application is ready for submission and not be concerned about a closing date.

Applications will be batched for assessment and sent to the most appropriate responsive mode panel meeting when they reach this stage of the process.

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4. Vision and Approach

Application question

What are you hoping to achieve with and how will you deliver your proposed work?

Create a document that includes your responses to all criteria. The document should not be more than six sides of A4, single spaced in paper in 11-point Arial (or equivalent sans serif font) with margins of at least 2cm. You may include images, graphs, tables. You can have an additional one page for a diagrammatic workplan.

Upload your response as a single PDF document, see details in the file upload guidance. Enter the words 'attachment supplied' in the text box.

For the Vision, explain how your proposed work:

- is of excellent quality and importance within or beyond the field(s) or area(s)
- has the potential to advance current understanding, generates new knowledge, thinking or discovery within or beyond the field or area
- is timely given current trends, context and needs
- impacts world-leading research, society, the economy or the environment

Within the Vision section we also expect you to:

- identify the potential direct or indirect benefits and who the beneficiaries might be

For the Approach, explain how you have designed your work so that it:

- is effective and appropriate to achieve your objectives
- is feasible, and comprehensively identifies any risks to delivery and how they will be managed
- if applicable, uses a clear and transparent methodology
- if applicable, summarises the previous work and describes how this will be built upon and progressed
- will maximise translation of outputs into outcomes and impacts
- describes how your, and if applicable your team's, research environment (in terms of the place, and relevance to the project) will contribute to the success of the work

Within the Approach section we also expect you to:

- demonstrate access to the appropriate services, facilities, infrastructure, or equipment to deliver the proposed work
- provide a project plan including milestones and timelines in the form of a Gantt chart or similar (additional one-page A4)

We attached our Vision and Approach

[NETBITS-Vision and Approach .pdf](#)

0.45MB

5. Applicant and team capability to deliver

Application question

Why are you the right individual or team to successfully deliver the proposed work?

Evidence of how you, and if relevant your team, have:

- the relevant experience (appropriate to career stage) to deliver the proposed work
- the right balance of skills and expertise to cover the proposed work
- the appropriate leadership and management skills to deliver the work and your approach to develop others
- contributed to developing a positive research environment and wider community

The word count for this section is 1,500 words, 1,000 words to be used for R4RI modules and, if necessary, a further 500 words for Additions.

Use the Résumé for Research and Innovation (R4RI) format to showcase the range of relevant skills you and, if relevant, your team (project and project co-leads, researchers, technicians, specialists, partners and so on) have and how this will help deliver the proposed work. You can include individuals' specific achievements but only choose past contributions that best evidence their ability to deliver this work.

Complete this section using the R4RI module headings listed. Use each heading once and include a response for the whole team, see the [UKRI guidance on R4RI](https://www.ukri.org/apply-for-funding/before-) (<https://www.ukri.org/apply-for-funding/before->

[you-apply/resume-for-research-and-innovation-r4ri-guidance/#:~:text=UKRI%20committed%20to%20adopting%20a,UKRI%20throughout%202022%20and%202023.](#)). You should consider how to balance your answer, and emphasise where appropriate the key skills each team member brings:

- contributions to the generation of new ideas, tools, methodologies, or knowledge
- the development of others and maintenance of effective working relationships
- contributions to the wider research and innovation community
- contributions to broader research or innovation users and audiences and towards wider societal benefit

Additions

Provide any further details relevant to your application. This section is optional and can be up to 500 words. You should not use it to describe additional skills, experiences, or outputs, but you can use it to describe any factors that provide context for the rest of your R4RI (for example, details of career breaks if you wish to disclose them).

Complete this as a narrative. Do not format it like a CV.

Within this section you can also demonstrate elements of your responses in visual form if relevant. You should:

- use images sparingly and only to convey important information that cannot easily be put into words
- insert each new image onto a new line
- provide a descriptive legend for each image immediately underneath it (this counts towards your word limit)
- ensure that files are smaller than 8MB and in JPEG, JPG, JPE, JFI, JIF, JFIF, PNG, GIF, BMP or WEBP format

UK Research and Innovation (UKRI) has introduced new role types for funding opportunities being run on the new UKRI Funding Service.

For full details, see [Eligibility as an individual](https://www.ukri.org/apply-for-funding/before-you-apply/check-if-you-are-eligible-for-research-and-innovation-funding/eligibility-as-an-individual/#contents-list) (<https://www.ukri.org/apply-for-funding/before-you-apply/check-if-you-are-eligible-for-research-and-innovation-funding/eligibility-as-an-individual/#contents-list>).

Team member	Initials	Short role descriptor
Ahmed Al-Dubai	AA	Lead PI of the project (ENU): Expertise in edge, VANETs, AI and Security. Will co-lead WP1, WP2 and WP3.
Berk Canberk	BC	Co-I (ENU) will lead WP1 as a leading expert in DT and will contribute to WP2.
Amir Hussain	AH	CO-I (ENU): Expertise in AI and equitable resource allocation for WP1 and WP2. Will co-lead WP1, WP2 and WP3.
Emma Hart	EH	CO-I (ENU: Expertise in multi-objective meta-heuristic based optimization methods. Will co-lead WP2 and oversee the optimization strategy in all WPs.
Achille Fonzone	AF	CO-I (ENU): Expertise in transport behaviour and the equity of the optimisation solutions. Will co-lead work on equity and optimisation (NEMO) in WP2
Baraq Ghaleb	BG	CO-I (ENU) will bring expertise in security, homomorphic encryption and blockchain. Will lead WP3
Tughrul Arslan	TA	Co-PI (UoE): Expert in intelligent power efficient sensing systems, resource allocation optimization and FPGA. Will work on DT in WP1, resource allocation in WP2, Privacy in WP3, and co-lead WP4.
Muhammad Imran	MI	Co-I (T4.1 Lead) (UoG): Expert in wireless and autonomous systems. Will oversee and co-lead WP4. He will also oversee the integration with 5G.
Olaoluwa Popoola	OP	Principal Investigator (WP4 Lead) (UoG) and will bring expertise in intelligent vehicles and blockchain. Will lead WP4
Lina Mohjazi	LM	Co-I (UoG) will bring expertise in performance evaluation, power transfer and machine learning. Will work on W4
Mahmoud Wagih	MW	Co-I (T4.2 Lead) (UoG); expertise in RF Energy Harvesting and wireless power transfer (WPT). Will co-lead the testbed development in WP4.

Module 1 – Contributions to the generation of new ideas

The team demonstrates a world-class track record in innovation, evidenced by their extensive portfolio of funded projects, highly cited works and numerous prestigious awards. **AA** led/co-led various projects funded by EPSRC, EU, World Bank and industry. He is the Co-PI of EPSRC COG-MHEAR programme. He received several best paper awards including IEEE Smart IoT 2019, IEEE IUCC 2015 and ACM MoMM 2013. **BC** led over 20 major projects and received several best paper awards in major conferences including the IEEE Flagship IEEE ICC 2024. **AH** is the PI of the EPSRC COG-MHEAR and other EPSRC projects, invited to join the Executive Committee of the UK Computing Research Committee (UKCRC) to promote digital inclusion. **EH** led several EPSRC, EU, and KTP-funded projects and her pioneering contribution to optimisation is recognised by the ACM SIGEVO Outstanding Evolutionary Computation Award. **AF**, an expert in transport analysis and user attitudes to automated vehicles; received Best paper award from the Scottish Transport Applications and Research conference 2016 and mobil.TUM 2016. **BG** focuses on secure computation and privacy, including work on homomorphic encryption and blockchain. **TA** led multiple EPSRC projects and holding over 20 patents. **MI** has 10 best paper awards in international conferences including IEEE ICC 2019. **OP**, an emerging expert in VANETs, developed the IS3C architecture for ultra-reliable 5G links, with work cited in IEEE standards. **MW**, head of Glasgow's GREEN RF Lab, created award-winning RF power transfer systems, leading to the RX Watt Ltd. spin-out. **LM**, developed analytical models to optimise energy-sustainable wireless networks.

Module 2 – The development of others

The team excels in fostering positive and productive research environments, demonstrating a deep commitment to mentorship, and the development of early-career researchers. **AA** leads research in his subject group of over 40 members, mentors several colleagues, and

has supervised successfully over 30 PhD and 200 MSc students. **BC** heads his Department and mentors early career staff. **AH** directs the Cognitive Big Data and Cybersecurity Research Lab at ENU, leading 30+ researchers and supervised 40 successful PhD students. **EH** is co-director of the Centre of AI and Robotics at ENU, leading over 30 academics, supervising 12 PhD students and mentoring researchers. **AF** emphasizes collaborative leadership, building professional growth environments, and supervises PhD students in different projects including the Horizon 2020 SUNRISE project. **BG** mentors early-career researchers and has supervised 8 PhD and 50+ MSc students. **TA** founded the Embedded Mobile, Wireless, and Wearable Sensor Systems group at UoE, leads System Level Integration, mentors colleagues, and supervises over 30 PhD students. **MI** is head of the School of Engineering at UoG, former Dean of Graduate Studies, and has guided over 1,000 graduate students and supervised 50+ PhD researchers. **OP** leads the £2M Glasgow 5G testbed, coordinates industry partners in the £5M EU Neutouch project and mentors postdocs, and PhD students. **MW** directs the £6M UKRI REACT Centre, managing staff across universities and partners, with prize-winning postdocs and students. **LM** mentors early-career researchers and PhD students. This collective experience ensures a supportive and high-performing environment for the PDRAs and staff on the NETBITS project.

Module 3 – Contributions to the wider research community

The team demonstrates a profound and sustained commitment to the wider research and innovation community through leadership roles in prestigious journals, conferences, and standards bodies. **AA** has delivered over 40 keynote talks globally, contributed to IETF IoT standards, chaired 41 international conferences/workshops, and associate editor for leading journals like IEEE Transactions on Sustainable Computing and ACM Computing Surveys. **BC** contributes through editorial roles and conference leadership. He is Associate Editor for IEEE Communications Surveys and Tutorials and delivered 30+ invited talks. **AH** is founding Editor-in-Chief of Cognitive Computation and multiple Springer series. He sits on editorial boards of leading AI journals, including IEEE Transactions on Artificial

Intelligence and Neural Networks and Learning Systems. **EH** has held leadership roles in the UK Operations Research Society, member of Scotland's AI Strategy steering committee; served on REF2021's Computer Science panel and was EiC of Evolutionary Computation (MIT Press) for 7 years. **AF** is Associated Editor of the Journal of Intelligent Transport Systems and member of the Editorial Board of Sustainability. **BG** supports research integrity, serving as Guest Editor and reviewer for leading journals. **TA** has served on IEEE committees, chaired key conferences like NASA/ESA AHS 2018, and delivered multiple keynotes. **MI** is the UK&RI Chair for the Backhaul/Fronthaul Networking & Communications (ETI-BNC) of IEEE ComSoc. and influenced policy through leadership roles (FIEEE, FIET, FRSE). **OP** serves on IEEE robotics committees. **MW** edits for IEEE Journal of Microwaves. **LM** leads IEEE gender diversity efforts in the UK and Ireland.

Module 4 – Contributions to broader research/innovation-users

The team has a strong record of translating research into tangible societal and economic benefits through impactful engagement with industry, policymakers, and the public. **AA** contributes to standardisation bodies (IEEE and IETF), and is regularly approached by media such as Aljazeera, Alhadath and Belqees TV to comment on cognitive IoT technologies. **BC** advises industry on digital twin technologies and leads impactful EU research projects. **AH** is a member of the Association for BME Engineers (ABFE-UK) to connect and inspire young professionals, women and other members of under-represented groups in computing and engineering. **EH** is an advocate for public engagement: her robotics TED talk has over 1.8 million views; her work has featured in the Guardian and New Scientist. **AF's** work addresses mobility equity, analysing travel behaviour and the cost-of-living crisis, and contributing to the CAV Forth autonomous bus project to inform inclusive transport policy. **BG** has delivered societal impact through AI and privacy research with partners like Leonardo MW Ltd. **TA's** innovations were adopted by major organisations such as NASA JPL, ESA, and Alpha Data. **MI**, a founding member of the Scotland 5G Centre, gave an invited TEDx talk (2015) and more than 50 plenary talks. **OP's** EPSRC-funded work on assistive

robotics (e.g., RoboGuide) has received national recognition, media coverage, and contributed to tech public engagements. **MW** translates wireless power technology into public benefit through spinouts, media (e.g., CBBC), and exhibits viewed by 10,000+ people. **LM** advises the UK Treasury and NISTA via the Royal Academy of Engineering.

Additional:

The project partners from Edinburgh Napier University (ENU), University of Edinburgh (UoE) and University of Glasgow (UoG) have already established a track-record of collaborations via working in joint funded projects such as EPSRC COG-MHEAR funded Programme (2021-2026), joint publications, joint postgraduate project supervision and internships. This creates mutual trust and team spirit that will enable the NETBITS project to operate effectively.

Managed from ENU with operations managers at UoE and UoG, NETBITS brings together a critical mass of researchers, supervisors, lecturers, mentors, industry partners, and regional authorities. The supervisory team comprising seven Professors, four early-career academics, five research associates, and two engineers offers perfectly matched expertise for the proposed research. Collectively, they have trained over 100 PhD students and research assistants, many now leading in academia and industry. The leadership team combines strong management capabilities with extensive experience in cohort-based training.

The consortium integrates the academic strengths of the three universities with the industrial expertise of CGI and AWTG Ltd. This partnership ensures the interdisciplinary depth required to transform intelligent communications and services for future-generation ITS. Industrial partners have committed in-kind support worth approximately £166K, ensuring exceptional value for money. By uniting academic and industry expertise, NETBITS will deliver innovative solutions that enhance the UK's competitive edge in high-tech economic sectors.

6. References

Application question

List the references you have used to support your application.

Include all references in this section, not in the rest of the application questions.

You should not include any other information in this section.

We advise you not to include hyperlinks, as assessors are not obliged to access the information they lead to or consider it in their assessment of your application.

If linking to web resources, to maintain the information's integrity, include persistent identifiers (such as digital object identifiers) where possible.

You must not include links to web resources to extend your application.

[1]. Y. Wu et al., "Intelligent Offloading Balance for Vehicular Edge Computing and Networks," in *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 5, pp. 5792–5803, May 2025.

[2]. Z. Ning et al., "Mobile Edge Computing and Machine Learning in the Internet of Unmanned Aerial Vehicles: A Survey," *ACM Comput. Surv.* 56, 1, Article 13, 31 pages. 2024

[3]. G. Walia, M. Kumar and S. S. Gill, "AI-Empowered Fog/Edge Resource Management for IoT Applications: A Comprehensive Review, Research Challenges, and Future Perspectives," in *IEEE Communications Surveys & Tutorials*, vol. 26, no. 1, pp. 619–669, First Quarter 2024.

[4]. X. Chen et al., "Vehicle as a Service (VaaS): Leverage Vehicles to Build Service Networks and Capabilities for Smart Cities," in *IEEE Communications Surveys & Tutorials*, doi:

10.1109/COMST.2024.3370169, 2024.

[5]. L. Zhao, A. **Al-Dubai**, A. **Hussain** et al. , "A Digital Twin-Assisted Intelligent Partial Offloading Approach for Vehicular Edge Computing," in *IEEE Journal on Selected Areas in Communications*, vol. 41, no. 11, pp. 3386–3400, Nov. 2023.

[6]. T. Do-Duy, B. **Canberk**, et al., "Digital Twin-Aided Intelligent Offloading with Edge Selection in Mobile Edge Computing," in *IEEE Wireless Communications Letters*, vol. 11, no. 4, pp. 806–810, April 2022, doi: 10.1109/LWC.2022.3146207

[7]. E. Ak, K. Duran, O. A. Dobre, T. Q. Duong and B. **Canberk**, "T6CONF: Digital Twin Networking Framework for IPv6-Enabled Net-Zero Smart Cities," in *IEEE Communications Magazine*, vol. 61, no. 3, pp. 36–42, March 2023, doi: 10.1109/MCOM.003.2200315

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[9]. M. S. Eddine, M. A. Ferrag, O. Friha, L. Maglaras, "EASBF: An efficient authentication scheme over blockchain for fog computing-enabled internet of vehicles," in *Journal of Information Security and Applications*, vol. 59, 2021,102802, doi: 10.1016/j.jisa.2021.102802.

[10]. A. Alabsi A. **Al-Dubai**, et al., "Wireless Power Transfer Technologies, Applications, and Future Trends: A Review," in *IEEE Transactions on Sustainable Computing*, vol. 10, no. 1, pp. 1–17, Jan.–Feb. 2025,

[11]. K Makanda, A. **Al-Dubai** et al., "Adaptive Mobile Chargers Scheduling Scheme Based on AHP-MCDM for WRSN," in *IEEE Transactions on Sustainable Computing*, vol. 10, no. 1, pp. 57–69, Jan.–Feb. 2025.

[12]. J. Zhang A. **Al-Dubai**, and A. Y. Zomaya, "A Novel Federated Learning Scheme for Generative Adversarial Networks," in *IEEE Transactions on Mobile Computing*, vol. 23, no. 5, pp. 3633–3649, May

2024, doi: 10.1109/TMC.2023.3278668.

[13]. Z. Li, **A. Al-Dubai** et al., "Reliable and Scalable Routing Under Hybrid SDVN Architecture: A Graph Learning Based Method," in *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 12, pp. 14022-14036, Dec. 2023, doi: 10.1109/TITS.2023.3300082

[14]. L. Zhao, **A. Al-Dubai**, et al., "MESON: A Mobility-aware Dependent Task Offloading Scheme for Urban Vehicular Edge Computing," in *IEEE Transactions on Mobile Computing*, vol. 23, no. 5, pp. 4259-4272, May 2024, doi: 10.1109/TMC.2023.3289611.

[15]. S. Peng, **A. Al-Dubai**, G. Min et al., "Secure Lightweight Stream Data Outsourcing for Internet of Things," in *IEEE Internet of Things Journal*, 8(13), July 2021.

[16]. A. Alsarhan, A. Itradat, A. Y. Al-Dubai, A. Y. Zomaya and G. Min, "Adaptive Resource Allocation and Provisioning in Multi-Service Cloud Environments," in *IEEE Transactions on Parallel and Distributed Systems*, vol. 29, no. 1, pp. 31-42, 1 Jan. 2018.

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[20]. A. Almaini, J. Folz, D. Woelfl, **A. Al-Dubai**, M. Schramm and M. Heigl, "A New Scalable Distributed Homomorphic Encryption Scheme for High Computational Complexity Models," *2023 International*

- Wireless Communications and Mobile Computing (IWCMC)*, Marrakesh, Morocco, 2023, pp. 890–897, doi: 10.1109/IWCMC58020.2023.10183131.
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- [22]. U. Anwar, Y. Dong, **T. Arslan**, K. Dashtipour, M. Gogate, **A. Hussain**, Q. Abbasi, **M. Imran**, T. Russ, P. Lomax "Privacy-Preserving Facial Emotion Classification with Visual Micro-Doppler Signatures for Hearing Aid Applications," in *IEEE Transactions on Instrumentation and Measurement*, vol. 74, pp. 1-10, 2025.
- [23]. M. Ferrag, M. Derdour, M. Mukherjee, A. Derhab, **L. Maglaras** and H. Janicke, "Blockchain Technologies for the Internet of Things: Research Issues and Challenges," in *IEEE Internet of Things Journal*, vol. 6, no. 2, pp. 2188–2204, April 2019, doi: 10.1109/JIOT.2018.2882794.
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- [26]. Dong, Z., Wu, H., Li, Z., Mi, D., **Popoola, O.** and Zhang, L., 2023, December. Trustworthy VANET: Hierarchical DAG-Based Blockchain Solution with Proof of Reputation Consensus Algorithm. In 2023 IEEE International Conference on Blockchain (pp. 127–132).
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10.1109/TIE.2023.3277112.

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[29]. Iturbe, X., Benkrid, K., Hong, C., Ebrahim, A., Torrego, R. & Arslan, T.; Microkernel Architecture and Hardware Abstraction Layer of a Reliable Reconfigurable Real-Time Operating System (R3TOS), *ACM Transactions on Reconfigurable Technology and Systems*, Feb 2015.

7. Project partners

Project partner	Contact	Direct contribution	Indirect contribution
CGI London , United Kingdom	Mattie Yeta mattie.yeta@cgi.com		£90,000 Access to lab, research time and shworcase, validate business model
AWTG Ltd London , United Kingdom	Ian Vernon ian.vernon@awtg.co.uk		£76,000 Access to labs, research time and testbed evaluation
DARe National Hub Glasgow, United Kingdom	David Flynn david.flynn@glasgow.ac.uk		£40,000 Provide advanced transport-modelling frameworks to assess NETBITS

8. Project partners: letters (or emails) of support

Application question

Upload a single PDF containing the letters or emails of support from each partner you named in the previous Project Partners section.

If you do not have any project partners, simply add 'N/A' into the text box, mark this section as complete and move to the next section.

If you have named project partners in the previous Project Partners section, enter the words 'attachment supplied' in the text box.

Each letter or email you provide should:

- confirm the partner's commitment to the project
- clearly explain the value, relevance and possible benefits of the work to them
- describe any additional value that they bring to the project
- refer to EPSRC's guidance on project partners letters of support for further information [Project partners letter of support – UKRI](https://www.ukri.org/councils/epsrc/guidance-for-applicants/what-to-include-in-your-proposal/project-partners-letter-of-support/) (<https://www.ukri.org/councils/epsrc/guidance-for-applicants/what-to-include-in-your-proposal/project-partners-letter-of-support/>)

Save letters or emails of support from each project partner you have listed in your application in a single PDF no bigger than 8MB. Unless specially requested, please do not include any personal data within the attachment.

Ensure you have prior agreement from project partners so that, if you are offered funding, they will support your project as indicated in the contributions template.

Do not provide letters of support from host and project co-leads' research organisations.

Please see the letters attached from CGI and AWGT Ltd.

[Letters of support APP41339.pdf](#)

0.41MB

9. Facilities

Application question

Does your proposed research require the support and use of a facility?

If you will need to use a facility, follow your proposed facility's normal access request procedures. Ensure you have prior agreement so that if you are offered funding, they will support the use of their facility on your project.

For each requested facility you will need to provide the:

- name of facility, copied and pasted from the [facility information list \(DOCX, 42KB\)](https://ukri-tfs-prod-assets.s3.eu-west-2.amazonaws.com/Facility+Information+for+TFS+updated+Sept+24.docx) (<https://ukri-tfs-prod-assets.s3.eu-west-2.amazonaws.com/Facility+Information+for+TFS+updated+Sept+24.docx>)
- proposed usage or costs, or costs per unit where indicated on the facility information list
- confirmation you have their agreement where required

Write 'N/A' if you do not need to use a facility.

N/A

10. Sensitive information

Application question

Is there sensitive information you need to share with UKRI that you do not want shared with assessors?

If you do not have anything to share, enter 'No' into the text box, mark this section as complete and move on to the next section.

If you, or a key team member, need to tell us something you wish to remain confidential, enter 'Yes' in the text box.

Then email EPSRC on tfschangeepsr@epsr.ukri.org You must

include in the subject line: <EPSRC responsive mode, sensitive info, Opportunity number, Application number>.

Typical examples of confidential information include:

- applicant is unavailable until a certain date (for example due to parental leave)
- declaration of interest
- additional information about eligibility to apply that would not be appropriately shared in the Applicant and team capability section
- conflict of interest for UKRI to consider in reviewer or panel participant selection
- the application is an invited resubmission

For information about how UKRI handles personal data, see [UKRI's privacy notice](https://www.ukri.org/about-us/privacy-notice/) (<https://www.ukri.org/about-us/privacy-notice/>).

NO

11. Resources and Cost

Total full economic costs (fEC)

£2,104,498

Total contribution from applying organisation(s)

£420,899.60

Total funding applied for

£1,683,598.40

Directly allocated

Cost categories

Category	Applied for
Staff	£208,108.00
Estates	£147,541.60

Other	£12,540.00
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Staff breakdown

Name	Role	%FTE
Olaoluwa Popoola	Project co-lead (UK)	10%
Lina Mohjazi	Project co-lead (UK)	5%
Muhammad Imran	Project co-lead (UK)	2%
Mahmoud Wagih	Project co-lead (UK)	5%
Technician	Technician	10%
Tughrul Arslan	Project co-lead (UK)	10%
Achille Fonzone	Project co-lead (UK)	4%
Baraq Ghaleb	Project co-lead (UK)	5%
Emma Hart	Project co-lead (UK)	3%
Amir Hussain	Project co-lead (UK)	5%
Berk Canberk	Project co-lead (UK)	5%
Ahmed Al-Dubai	Project lead	15%

Directly incurred

Cost categories

Category	Applied for
Staff	£553,644.00
Equipment	£2,880.00
Travel and subsistence	£37,120.00
Other	£73,632.00

Staff breakdown

Name	Role	%FTE
RA (18m)	Research and innovation associate	100%
RA (12m)	Research and innovation associate	100%
PDRA	Research and innovation associate	100%
Research Assistant	Research and innovation associate	100%
Research Associate	Research and innovation associate	100%

Indirect

Cost categories

Category	Applied for
Indirect costs	£648,132.80

Staff breakdown

No staff for this category

Exceptions

Cost categories

Category	Applied for
Staff	-

Equipment	-
Travel and subsistence	-
Other	-

Staff breakdown

No staff for this category

[Full cost breakdown by organisation \(/applications/APP41339/resources-and-costs/detailed-view?](/applications/APP41339/resources-and-costs/detailed-view?backlinkHref=%2Fapplications%2FAPP41339%3Ftab%3DreadTab&backlinkTitle=Back+to+Application)

[backlinkHref=%2Fapplications%2FAPP41339%3Ftab%3DreadTab&backlinkTitle=Back+to+Application\)](/applications/APP41339/resources-and-costs/detailed-view?backlinkHref=%2Fapplications%2FAPP41339%3Ftab%3DreadTab&backlinkTitle=Back+to+Application)

Justification of resources

JUSTIFICATION of RESOURCES (JoR) AND MANAGEMENT

(WP5), Management: The Project Management Committee (PMC), chaired by the PI and comprising all investigators, will meet monthly to monitor progress, assess risks, and take strategic decisions. Staff will provide weekly updates, while the PMC oversees resources, dissemination, and IPR agreements prior to project launch. Quarterly all-hands meetings with industrial partners, weekly team discussions across ENU, UoE, and UoG, and regular video conferencing will ensure integration and effective communication. Supervision will include one-to-one reviews, bi-monthly coordination, site visits, and web-based activities. ENU will host a secure project website with public results, software, and outreach materials. Researchers will also spend two weeks annually with partners CGI, DARE Hub, and AWTG Ltd to reinforce collaboration.

Research Infrastructure and environment:

Edinburgh Napier University (ENU), University of Edinburgh (UoE), and University of Glasgow (UoG) collectively provide an outstanding research environment that integrates advanced technological

infrastructures, interdisciplinary collaboration, and strong institutional support. ENU will contribute its expertise in AI-driven edge vehicular communication, security, and Digital Twin-as-a-Service (DTaaS) for high-frequency urban mobility, transport policy, and engineering across WPs 1–3. UoG will lead the modelling and validation of the proposed platform, supported by state-of-the-art 5G-enabled wireless charging testbeds and blockchain-based mobility infrastructures aligned with national decarbonisation objectives (WPs 3–4). UoE offers distinctive capabilities in AI-enabled heterogeneous FPGAs, enabling real-time decision support and computationally intensive models within WP4. Collectively, these institutions cultivate a dynamic ecosystem defined by innovation, sustainability, and interdisciplinary research. The partnership unites experts in AI, digital twins, intelligent transport systems, optimisation, security, embedded systems, and wireless power transfer. This synergistic environment not only advances academic excellence but also delivers tangible societal impact through transformative solutions and strategic partnerships.

Staff – Directly Allocated Posts

ENU: Prof. A. Al-Dubai (Lead PI) will manage the PM C, co-lead WP1, WP3, and WP4, contributing 15% of his time to ensure project integration. He will primarily supervise ENU PDRA1 and PDRA2, and prepare reports, findings, and deliverables. Prof. A. Hussain (Co-PI) will contribute 5% of his time to co-lead WP2 and WP3, support AI work in other WPs, and prepare reports and papers. Prof. B. Canberk will commit 5% of his time to lead WP1 and contribute to WP4. Prof. E. Hart (CI) will contribute 3% of her time to WP2, leading work on proactive resource allocation strategies, evolutionary algorithms, and optimisation, and preparing reports and academic outputs. Prof. A. Fonzone (CI) will commit 3% of his time to the innovative Net-zero Equitable Mobility Optimisation (NEMO) work. Dr. B. Ghaleb will commit 5% of his time to lead WP3. ENU will offer a PhD studentship to support the project.

UoG: Dr. O. Popoola will lead WP4, co-lead WP1 and WP2, contributing 10% of his time. He will supervise UoG PDRA3, liaise with industry

partners on the wireless charging as a service model, prepare academic papers, and manage dissemination and standardisation. Dr. M. Wagih will commit 5% of his time to co-lead WP4, supervising UoG PDRA4 in developing the wireless charging testbed. Prof. M. Imran will contribute 2% of his time to develop an intelligent knowledge management framework for WP4. Dr. L. Mohjazi will contribute 5% of her time to WP2 and WP4.

UoE: Prof. Arslan will commit 10% of his time to co-lead the project, contribute to DT in WP1, Cognitive Engine and Multi-objective Optimisation in WP2, and FPGA work in WP4. He will supervise project PDRAs and lead dissemination of project results.

Staff – Directly Incurred Posts

PDRA1 (ENU) 36 Months: will collaboratively identify and address technical requirements relating to development of DT and synchronisation solutions in (WP1). PDRA1 will also contribute to the required scenarios to integrate and validate proposed (WP1) solutions in WP4. PDRA1 will lead the development and integration of all deliverables in WP1 and WP2.

PDRA2 (ENU) 36 Months: will address the edge, optimisation in WP1 and WP2 and secure data access control and privacy-preserving [WP3] and will implement the security solutions in WP4. The aim is to develop trust model to ensure the security and privacy of the V2V and V2X communications. The model will manage the entire architecture developed in WP1 considering both the service level and the system level. PDRA2 will liaise with PDRA1 and PDRA3 and the rest of the team for evaluation of the developed solutions in WP3 integration in WP4.

PDRA3 (Glasgow) 18 Months: will investigate real time wireless charging in dynamic vehicular scenarios. PDRA3 will define and optimize the problem of wireless charging in a connected vehicular environment and deploy his solution on the edge servers at UoG 5G testbed and integrate the developed solution to a bespoke wireless charging testbed for experimental validation of the models developed.

PDRA4 (Glasgow) 12 Months: will focus on development of an automated EV wireless charging testbed using the 6.78 MHz Airfuel™ standard, featuring multi-agent simulations, high-efficiency power electronics, and real-time data acquisition. He/she will work with PDRA3 to integrate the testbed to UoG a 5G testbed and facilitate the wireless charging database creation.

PDRA5 (Edinburgh) 36 Months: will investigate the Synchronisation Optimisation of the DT in WP1, the access control in WP3 and will work on the FPGA and Distributed AI-enabled DT Testbed in WP4.

Travel & Subsistence (£89,920)

To allow the three academic partners present findings in leading international conferences and workshops. This including Meetings, kick-off, final workshop, annual general assemblies and enable PDRAs to spend 2 weeks a year at industrial partners.

Equipment/testbed (£48,521)

Electronic components for the wireless charging and mini Robots / autonomous vehicles. The University of Glasgow's 5G testbed enables low-latency, high-reliability testing of vehicular communication, AI optimisation, blockchain-secured data, and wireless charging. Strategic scheduling condenses experiments into 65 days over three years, cutting costs from £465,802 to £28,060 while preserving full access, ensuring rigorous, reproducible validation and maximising value for NeTBITS.

In-Kind Contributions (£206K)

CGI Ltd: £90K, AWTG Ltd: ~£76K and DARE National Hub: £40K.

This ensures efficient balance between academic research and industrial collaboration, enabling NeTBITS' transformative goals while partners collaboratively define requirements for use cases in WP4.

12. Ethics and responsible research and innovation (RRI)

Application question

What are the ethical or RRI implications and issues relating to the proposed work? If you do not think that the proposed work raises any ethical or RRI issues, explain why.

Demonstrate that you have identified and evaluated:

- the relevant ethical or responsible research and innovation considerations
- how you will manage these considerations

Within this section you can also demonstrate elements of your responses in visual form if relevant. You should:

- use images sparingly and only to convey important information that cannot easily be put into words
- insert each new image onto a new line
- provide a descriptive legend for each image immediately underneath it (this counts towards your word limit)
- ensure files are smaller than 8MB and in JPEG, JPG, JPE, JFI, JIF, JFIF, PNG, GIF, BMP or WEBP format

No ethical or RRI implications as the work is mainly in developing algorithms with no ethical concerns.

13. Genetic and biological risk

Application question

Does your proposed research involve any genetic or biological risk?

In respect of animals, plants or microbes, are you proposing to:

- use genetic modification as an experimental tool, like studying gene function in a genetically modified organism
- release genetically modified organisms
- ultimately develop commercial and industrial genetically modified outcomes

If yes, provide the name of any required approving body and state if approval is already in place. If it is not, provide an indicative timeframe for obtaining the required approval.

Identify the organism or organisms as a plant, animal or microbe and specify the species and which of the three categories the research relates to.

Identify the genetic and biological risks resulting from the proposed research, their implications, and any mitigation you plan on taking. Assessors will want to know you have considered the risks and their implications to justify that any identified risks do not outweigh any benefits of the proposed research.

Write 'N/A' if this does not apply to your proposed work.

No genetic and biological risk

14. Research involving the use of animals

Application question

Does your proposed research involve the use of vertebrate animals or other organisms covered by the Animals Scientific Procedures Act?

If you are proposing research that requires using animals, download and complete the [Animals Scientific Procedures Act template \(DOCX, 74KB\)](https://ukri-tfs-prod-assets.s3.eu-west-2.amazonaws.com/AnimalResearchQuestionTemplate-BBSRC.docx) (<https://ukri-tfs-prod-assets.s3.eu-west-2.amazonaws.com/AnimalResearchQuestionTemplate-BBSRC.docx>), which

contains all the questions relating to research using vertebrate animals or other Animals (Scientific Procedures) Act 1986 regulated organisms. Then, save it as a PDF.

Write 'N/A' if this does not apply to your proposed work.

No animals involved

15. Conducting research with animals overseas

Application question

Will any of the proposed animal research be conducted overseas?

If you are proposing to conduct overseas research, it must be conducted in accordance with welfare standards consistent with those in the UK, as in Responsibility in the use of animals in bioscience research, page 14 (<https://www.nc3rs.org.uk/3rs-resources/responsibility-use-animals-bioscience-research>).

Ensure all named applicants in the UK and overseas are aware of this requirement. Provide a statement to confirm that:

- all named applicants are aware of the requirements and have agreed to abide by them
- this overseas research will be conducted in accordance with welfare standards consistent with the principles of UK legislation
- the expectation set out in Responsibility in the use of animals in bioscience research will be applied and maintained
- appropriate national and institutional approvals are in place

Overseas studies proposing to use non-human primates, cats, dogs, equines or pigs will be assessed during NC3Rs review (<https://www.nc3rs.org.uk/who-we-are-and-what-we-do#services>) of research applications. Provide the required information by completing the template from the question 'Research involving the use of

animals’.

For studies involving other species, select, download, and complete the relevant Word checklist or checklists from this list:

- Additional questions on the use of rodents overseas
(https://nc3rs.org.uk/sites/default/files/documents/Additional%20questions%20on%20the%20use%20of%20rodents%20overseas%20%286%29_0.docx)
- Additional questions on the use of rabbits overseas
(<https://nc3rs.org.uk/sites/default/files/documents/Additional%20questions%20on%20the%20use%20of%20rabbits%20overseas.docx>)
- Additional questions on the use of sheep overseas
(<https://nc3rs.org.uk/sites/default/files/documents/Additional%20questions%20on%20the%20use%20of%20sheep%20overseas.docx>)
- Additional questions on the use of goats overseas
(<https://nc3rs.org.uk/sites/default/files/documents/Additional%20questions%20on%20the%20use%20of%20goats%20overseas.docx>)
- Additional questions on the use of pigs overseas
(<https://nc3rs.org.uk/sites/default/files/documents/Additional%20questions%20on%20the%20use%20of%20pigs%20overseas.docx>)
- Additional questions on the use of cattle overseas
(<https://nc3rs.org.uk/sites/default/files/2021-12/Additional%20questions%20on%20the%20use%20of%20cattle%20overseas.docx>)
- Additional questions on the use of *Xenopus laevis* and *Xenopus tropicalis* overseas
(<https://nc3rs.org.uk/sites/default/files/Additional%20questions%20on%20the%20use%20of%20Xenopus%20laevis%20%26%20Xenopus%20tropicalis%20overseas.docx>)

Save as a PDF. If you use more than one checklist, save it as a single PDF.

Write 'N/A' if this does not apply to your proposed work.

No

16. Research involving human participation

Application question

Will the project involve the use of human subjects or their personal information?

If you are proposing research that requires the involvement of human subjects, provide the name of any required approving body and whether approval is already in place.

Justify the number and the diversity of the participants involved, as well as any procedures.

Provide details of any areas of substantial or moderate severity of impact.

Write 'N/A' if this does not apply to your proposed work.

No

17. Research involving human tissues or biological samples

Application question

Does your proposed research involve the use of human tissues, or biological samples?

If you are proposing work that involves human tissues or biological samples, provide the name of any required approving body and whether approval is already in place.

Justify the use of human tissue or biological samples specifying the nature and quantity of the material to be used and its source.

Write 'N/A' if this does not apply to your proposed work.

NO
